



Automatic varied-length ECG classification using a lightweight DenseNet model

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ABSTRACT

Atrial fibrillation is the most common abnormal heart condition and contributes primarily to cardiac morbidity and mortality. In the last decades, portable Electrocardiogram (ECG) monitor has effectively supported patients with periodic heart arrhythmia by constantly monitoring heart activity and early detecting heart arrhythmia. However, diagnosing the ECG recordings, which is a demanding and time-consuming job, is mainly relied on experienced medical workers. It is of great importance to develop methods to assist in diagnosing ECG recordings. In this work, we propose a lightweight model architecture based on the Densely Connected Convolutional Networks to classify cardiac arrhythmia automatically. The Cyclical Learning Rate was utilized to automatically change the learning rate value throughout the training procedure. Our model was trained and tested on the PhysioNet/Computing in Cardiology Challenge 2017 (CinC2017) and 1st China Physiological Signal Challenge 2018 (ICBEB2018) datasets. The obtained results have been compared with other works using metric F_1 for CinC2017 and metrics $F_1, AUC, F_{max}, F_{\beta=2}, G_{\beta=2}$ for ICBEB2018. Though our structure consists of a low number of parameters and requires less computational costs, its performance is in line with the state-of-the-art Deep Neural Networks with F_1 score of 0.831 and 0.826 for CinC2017 and ICBEB2018 datasets, respectively.

1. Introduction

Atrial fibrillation (AF) is a disorder of the electrical currents in the atria that causes muscle twitches rapidly and irregularly. The main reason for AF is electricity flows back from the ventricles, or electricity is blocked and bounced back. It is the most common heart problem and contributes primarily to cardiac morbidity and mortality [1]. Therefore, it is greatly important to detect AF early in order to provide patients with timely care and treatment. In the last decades, the advent of portable monitoring devices, which can be used to constantly monitor heart activity, has effectively supported patients with periodic heart arrhythmia. However, these devices cannot classify the ECG signals, but it is the job of physicians, nurses, or other medical workers. Unfortunately, it is not trivial to interpret the ECG recordings, and it significantly depends on the knowledge and experience of medical workers. Hence, it is expected to be mistakenly analyzed and diagnosed sometimes [2,3]. To overcome this difficulty, there have been many intensive studies on developing reliable methods that allow an automatic ECG signal classification.

To inspire the development of approaches for automatically classifying arrhythmia from ECG signals, in 2017 PhysioNet organized

an annual challenge to identify AF symptoms from short single-lead ECG signals [4]. Four teams that share first place in this challenge use different approaches, including hand-engineered features fed to a random forest, extreme gradient boosting, Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs) [5–8]. Next to the 2017 PhysioNet/Computing in Cardiology Challenge, the 1st China Physiological signal challenge took place in 2018. This challenge is to encourage the development of methods to detect the morphology or rhythm abnormalities from multi-lead ECG signals [9]. The team of Tsai-Min Chen et al. used a convolution neural network to classify ECG signals and won this challenge with the F_1 score of 0.84 [10].

Outside the scope of the above challenges, many methods and models have been developed to automatically identify AF rhythm and proven to work relatively well. Based on the fact that the absence of P waves and irregular R–R intervals are characteristic of AF rhythm, many methods have been developed to detect AF by observing P waves absence [11–13], or irregular R–R intervals [14–19], or both [20–22]. As the P wave is weak and sensitive to noise, detecting its abnormality is difficult. However, when atrial fibrillation occurs, the irregularity of the R–R interval can be observed by the ECG signals regardless

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of the noise effect. Therefore, the atrial fibrillation detection methods based on R-R interval perform better than those based on P wave [23]. Recently, the methods based on deep learning algorithms and CNNs have been widely utilized for ECG signal diagnosis [24–26]. In [24], one-dimensional ECG signals were segmented and converted to two-dimensional images using a recurrence plot which were subsequently used as inputs to the CNN models. A combination of CNN and Modified Frequency Slice Wavelet Transform (MFSWT) was used to detect AF from a short-term episode of ECG signal in the work of Xu et al.. They obtained a mean accuracy of 81.07% on the MIT-BIH dataset, and that value increases to 84.85% when excluding an extremely low quality ECG signal [25]. Using the same dataset, Xia et al. first apply Stationary Wavelet Transform (SWT) and Short-Term Fourier Transform (STFT) for signal conversion, then they develop two corresponding deep convolutional neural network models. Their method shows high accuracy, sensitivity, and specificity [26]. In the literature [25,26], raw ECG data were transformed into suitable representation as inputs for the models by feature engineering-based techniques (e.g., wavelet transform). Unfortunately, these algorithms are susceptible to interference in ambulatory devices and have relatively low generalization on new data [19]. Other than CNNs, recurrent neural network architectures were proven to be suitable for classifying ECG recordings as well [27]. More recently, transfer learning technique has been used to share knowledge between related tasks and utilize the commonalities between different ECG conditions [28–30]. It was shown that with the transfer learning technique, the model does not require a large number of expensive annotations but still outperforms CNNs that are not pretrained, especially when the train set is not sufficient.

Compared with other model architectures, the Densely Connected Convolutional Networks (DenseNets) is an architecture in which each layer is connected directly to all subsequent layers. That allows for a reduction of the number of parameters of the network and is promising for classifying ECG signals with a suitable real-time implementation. In this work, we propose a lightweight network architecture based on DenseNets [31] with several modifications to classify cardiac arrhythmia automatically. In addition, the Cyclical Learning Rate (CLR) [32] was also used to change the learning rate hyperparameter value throughout the training procedure automatically. The performance of our model is in line with the state-of-the-art models even though it consists of a small number of parameters; therefore, this model can be easily used and further developed with a low computational cost. We verify the accuracy as well as the robustness of the proposed model and compare the results obtained with other works on CinC2017 and ICBEB2018 datasets.

2. Materials and methods

2.1. Materials

We use two freely accessible databases, which were collected to inspire the development of approaches to identify cardiac arrhythmia from short ECG signals, to train and evaluate our model. The first one is the 2017 Physionet/CinC Challenge (CinC2017) dataset [4]. This dataset consists of 12186 single-lead ECG recordings in total with 8528 of them are in the public training set, and the remaining 3658 recordings are in the private hidden test set. Each recording lasts 9 to 61 s, performed by patients, and has one label (AF rhythm, normal rhythm, other rhythm, and noisy type). The data were stored as 300 Hz, 16-bit recordings with a ± 5 mV dynamic range and a bandwidth of 0.5–40 Hz. The second dataset is from the 1st China Physiological Signal 2018 (ICBEB2018) [9]. This dataset includes 6877 standard 12-lead ECG records collected from 11 hospitals and from both male and female patients. These ECG records are from 6 s to 60 s and sampled at a frequency of 500 Hz. Each recording has at least one label and at most three labels that indicate heart conditions. There are 9 labels in total, including normal sinus rhythm and 8 types of arrhythmia (AF,

PAC, PVC, STD, STE, I-AVB, LBBB, and RBBB). Similar to the CinC2017 dataset, the test set of the ICBEB2018 dataset is not public either.

Since the signals in both databases have different lengths, to uniform the input data lengths, the short signals have been extended by adding zero-valued data to the beginning of the signals. For the CinC2017 dataset, the sampling frequency of the signals is preserved, and the length of the signals is extended to 60 s. In the case of the ICBEB2018 dataset, due to Google Colab Pro's limited RAM capacity, we reduced the sampling frequency of the signal to 250 Hz and the signal length only extended to 30 s. In addition, the data is also normalized using mean and standard deviation calculated over the entire dataset before being fed into the model for training.

2.2. Methods

2.2.1. Proposed lightweight DenseNet architecture

In this study, we used Densely Connected Convolutional Networks (DenseNets), which is originally developed by Gao Huang et al. [31], to classify cardiac arrhythmias automatically. Unlike the traditional convolutional networks where each layer is connected to only one subsequent layer. In DenseNets, each layer is connected directly to all subsequent layers. By doing so, the feature maps of all preceding layers are transferred into each layer; therefore, fewer parameters are needed to achieve equally well performance compared to traditional convolutional networks. In the original work, DenseNets architecture is designed for two-dimensional image classification purposes and tends to use small 3×3 convolutional kernels to identify significant regional variations. Architecturally, our network follows the original DenseNet-B structure version with several small modifications to suit our problem. First of all, our network uses only three dense blocks with 6, 4, and 6 convolutional layers, respectively. Second, since the input ECG signal is one-dimensional data, the two-dimensional convolutional layers have been replaced by one-dimensional convolutional layers. Third, as the ECG is a low-frequency biomedical signal that is susceptible to noise and is sampled at a low rate, it is difficult to use convolutional kernels with a small size of 3×3 to shape the meaningful changing signal regions. Therefore, we have used larger-sized convolution kernels, such as 8, 6, and 8 at each stage, respectively, to deal with this problem. These suggested larger-size convolution kernels were observed to work better than the original 3×3 filters. The growth rate and the number of channels in the input layer, which are architectural hyperparameters, are chosen to be 8 and 16, respectively. Thus, our network consists of 37 convolutional layers and is denoted by DenseNet-37. Although the number of parameters of the proposed model is significantly smaller than other DenseNet models, it allows us to solve the same problem with less computational costs while obtaining comparable or even better accuracy. That is why we call our model lightweight. The architecture of the proposed DenseNet-37 model and the number of parameters of different DenseNet models are presented in Fig. 1 and Table 4, respectively.

2.2.2. Model training

We trained and evaluated the CNN models on the CinC2017 and ICBEB2018 datasets, which represent two common classification problem types, including single-label and multi-label classification tasks. In the CinC2017 dataset, each ECG signal has one label; hence, we chose soft-max as the activation function for the full connected layer and categorical cross-entropy as loss function. In contrast, each ECG signal of the ICBEB2018 dataset has up to three labels, so we use sigmoid and binary cross-entropy as the activation function and loss function to train our network. The dense layer with output matching the number of classes of each dataset employed to classify the patterns. To maximize generalization, the weight regularization technique with $l1 = 1e-5$ and $l2 = 1e-5$ is used.

The selection of the optimization algorithm for training the deep neural networks is of great importance and it affects the process of

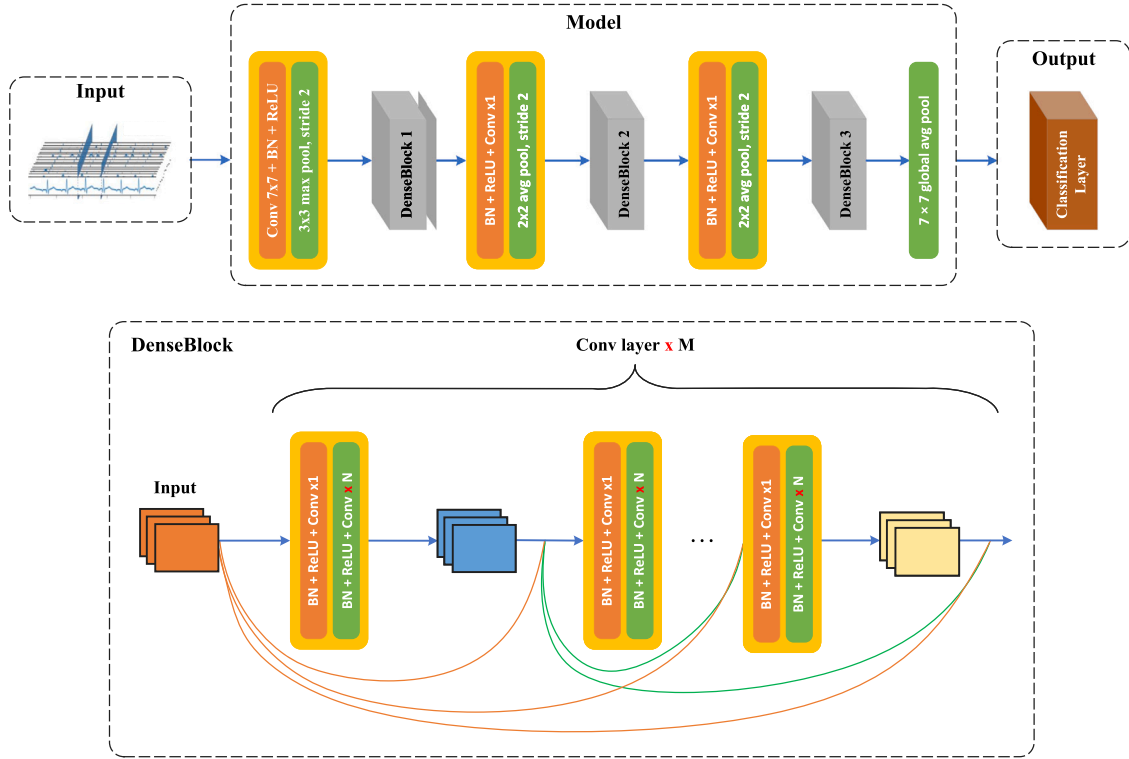


Fig. 1. Framework of the proposed DenseNet-37 model, which contains three dense blocks. Each block has 6, 4, and 6 convolution layers (denoted as M in the figure), with the convolution kernel of 8, 6 and 8 (denoted as N in the figure), respectively.

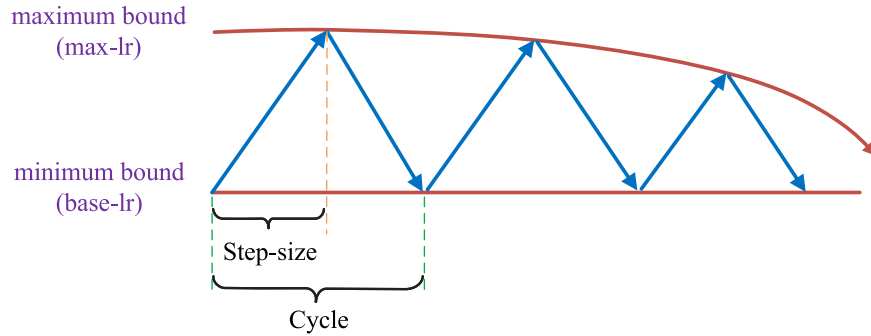


Fig. 2. The "exp_range" cyclical learning rate policy. The learning rate values (blue lines) changing between minimum bound and maximum bound. Step-size is the number of iteration in half a cycle.

optimizing the weights as well as the model's classification results. In this study, we deploy the Adam optimizer for deep experiments. We use the following settings: initial lr = $1e-5$, decay = 0.0, epsilon = $1e-9$, beta₁ = 0.9, beta₂ = 0.98 as the initialization of parameters for the optimizer. One of the hyperparameters that significantly affect the training process of deep CNNs is the learning rate. Usually, this parameter has a large value in the first epochs but gradually decreases in the later epochs. In our work, we initialize the learning rate for the Adam optimizer to $1e-4$, then the Cyclical Learning Rate (CLR) technique [32] was used to automatically change the learning rate value from base lr = $1e-5$ (minimum bound) to max lr = $1e-3$ (maximum bound) cyclically. By doing so, we achieve improved classification performance without the need to tune and to converge in fewer iterations. To give more fine-tuned control in the rate of decline in maximum bound of the learning rate, we used exp-range policy after each cycle step_size = 8 epochs as presented in Fig. 2.

Furthermore, the early stopping approach was also used with minimum change in loss value as $1e-3$ and the amount of patience as 30 to avoid over-fitting. The training procedure will be interrupted if the

validation loss value stops improving after 30 epochs. During training, macro value of the desired score on the validation set will be recorded after each training epoch. When the training is done, the weights of the trained model will be reverted to the checkpoint at which the network had the best score on the validation set. Finally, the model with the weights obtained will be evaluated on the test set and the desired score is recorded. All networks use a validation step of 1.

As mentioned previously, the test sets of two databases used in this work are kept private. To investigate the classification ability of the CNN networks in the deep experiments, the datasets were randomly split into test, validation, and training sets at 20%, 5%, and 75% of recordings, respectively. The training and evaluating procedure are repeated 10 times, each time the test set is kept the same while the validation and training sets are taken from a pool of 80% with the same class ratio in each set. Finally, the identification ability of a network was assessed based on the average macro desired scores and the standard deviation on the test set after 10 runs. The obtained results are reported in Tables 1–6 as $M (\pm \text{STD})$, where M and STD

Table 1

The F_1 scores achieved depending on the mini-batch size for the CinC2017 dataset [4] on our test set. The score in each cell is presented as the average macro value ($\pm$ standard deviation). The score with the greatest average value in each column is bolded.

Mini-batch size	Model performance				
	F_1	F_{1n}	F_{1a}	F_{1o}	F_{1p}
8	.791 ($\pm$.009)	.916 ($\pm$.003)	.799 ($\pm$.020)	.779 ($\pm$.011)	.668 ($\pm$.024)
16	.779 ($\pm$.010)	.913 ($\pm$.005)	.791 ($\pm$.018)	.772 ($\pm$.013)	.639 ($\pm$.028)
32	.770 ($\pm$.015)	.902 ($\pm$.008)	.768 ($\pm$.025)	.743 ($\pm$.020)	.668 ($\pm$.024)
64	.759 ($\pm$.017)	.901 ($\pm$.007)	.744 ($\pm$.034)	.749 ($\pm$.018)	.641 ($\pm$.026)
128	.736 ($\pm$.011)	.879 ($\pm$.017)	.715 ($\pm$.026)	.685 ($\pm$.021)	.665 ($\pm$.030)

are respectively the average macro desired score and the standard deviation.

3. Results and discussion

We first describe metrics that are used in this work to evaluate model performance. In the next step, we test how our model performs with different mini-batch and training set sizes. Then we verify the accuracy of the proposed model by comparing the results achieved with other works using different metrics. Finally, the robustness of our model is demonstrated when it works well for different training sets and for both single-label (CinC2017 dataset) and multi-label (ICBEB2018 dataset) classification tasks.

3.1. Performance metrics

The classification ability of our trained model is evaluated based on an F_1 measure, which is the average F_1 value from all classification types (4 types with CinC2017 dataset, and 9 types with ICBEB2018 dataset). The detail calculation of F_1 scores of each class is described in Ref. [4,9]. In addition, the efficiency of the proposed model architecture was also assessed for ICBEB2018 dataset using other four metrics: (i) the averaged class-wise AUC (AUC), (ii) the maximum F_1 score found by varying the decision threshold, F_{max} ; (iii) a general class-weighted F-score F_β ; and (iv) a generalization of the Jaccard measure G_β . For both F_β and G_β , the sample weights are inversely proportional to the number of labels and they are calculated using formulas (1) and (2) with $\beta = 2$, respectively. The first two metrics were following the work of Strodthoff et al. [30] and the last two were used in the PhysioNet/CinC Challenge 2020.

$$F_\beta = \frac{(1 + \beta^2).TP}{FP + \beta^2.FN + (1 + \beta^2).TP} \quad (1)$$

$$G_\beta = \frac{TP}{TP + FP + \beta.FN} \quad (2)$$

where True Positive (TP) and True Negative (TN) are the number of positive and negative samples that were also predicted as positive and negative respectively by the model. False Positive (FP) and False Negative (FN) are the number of negative and positive samples that were incorrectly predicted as positive and negative respectively by the model.

3.2. Experimental results

The metrics achieved by the proposed DenseNet-37 network tested on the PhysioNet/CinC Challenge 2017 [4] with the different mini-batch sizes are presented in Table 1. F_1 is the average value of the score of four classification types, including the normal rhythm (F_{1n}), AF rhythm (F_{1a}), other rhythm (F_{1o}), and noisy type (F_{1p}). In this experiment, the dataset was randomly divided into validation (5%), training (75%), and test (20%) sets. As shown Table 1, the value of F_{1p} is lowest among four classification types, ranging from 0.639 to 0.668. Small F_{1p} value leads to an overall lower F_1 score compared to the case when F_1 is calculated as average value of F_{1n} , F_{1a} , and F_{1o} only. All F_1

Table 2

Investigation of the F_1 score depending on the size of the training set for the CinC2017 dataset [4]. The score in each cell is presented as the average macro value ($\pm$ standard deviation). The score with the greatest average value in each column is bolded.

Dataset	Model performance				
	F_1	F_{1n}	F_{1a}	F_{1o}	F_{1p}
DS1	.791 ($\pm$.009)	.916 ($\pm$.003)	.799 ($\pm$.020)	.779 ($\pm$.011)	.668 ($\pm$.024)
DS2	.771 ($\pm$.010)	.902 ($\pm$.005)	.778 ($\pm$.014)	.737 ($\pm$.013)	.667 ($\pm$.038)
DS3	.718 ($\pm$.017)	.808 ($\pm$.008)	.710 ($\pm$.033)	.664 ($\pm$.031)	.619 ($\pm$.045)

Different datasets (DS) in the table were generated with the training-validation-test ratio as follows: (1) 75%:5%:20% (DS1); (2) 50%:5%:20% (DS2) and (3) 25%:5%:20% (DS3).

scores are highest with the mini-batch size of 8, and F_1 scores slowly decrease when the mini-batch size increases.

To examine the robustness of the proposed model, we use the different training and validation sets while the test set is kept the same. Comparison of the F_1 score of the different training set sizes for the CinC2017 dataset is shown in Table 2. It is obvious that larger training sets provide better accuracy. When the training set reduces from 75% to 25%, the F_1 slightly decreases from 0.791 to 0.718. However, the reduction is not dramatic, which implies that the proposed model works relatively well even with a training set of small sizes.

In Table 3, the F_1 scores achieved by different training methods in the experiments for the CinC2017 dataset are shown. In Refs. [33–38], F_1 is taken as the average score value of three classification types (F_{1n} , F_{1a} , F_{1o}), and F_{1p} is not taken into account. If we do the same, the F_1 value from our model will be 0.831. Our model performs comparably well compared to other state-of-the-art works using the same metric while requiring less computational cost and less complexity. In Ref. [33], authors have access to the hidden test set of the CinC2017 dataset; therefore, they can use all 8528 single-lead ECG recordings for training. Additionally, this study applied Discrete Wavelet Transform (DWT) to remove noise before performing the ECG classification. That should be one of the reasons why their results of F_1 score is relatively high compared to other studies. In the work of Fan et al. the U-Net model was used to determine the R peaks, and R–R distances [35]. The input data of the U-Net model has been augmented to balance different classes, and the sampling frequency of the ECG signal was kept unchanged at 500 Hz. In the next step, a 34-layer residual network was utilized to capture waveform morphological features from ECG signals. Finally, the ECG classification was performed to identify the abnormal rhythm types. As a result of a sophisticated procedure, the obtained F_1 score is the best among all up-to-date methods.

To check the ability of our model in multi-label classification task, we use it for the ICBEB2018 dataset, in which each ECG recording has up to three labels. Table 4 and Table S1 show the F_1 scores achieved by the different DenseNets model architectures. Similar to the CinC2017 dataset, we randomly split the ICBEB2018 dataset into training, validation, and test sets with the ratio of 75% – 5% – 20%. The mini-batch size of 8 is used as it was proven to be better than the others previously. As shown in these tables, the proposed DenseNet-37 outperforms all original DenseNets, which was developed by Gao Huang et al. [31]. DenseNet-37 not only performs the best but also has

Table 3

Comparison of the F_1 scores achieved by different works for the CinC2017 dataset [4]. The score in each cell is presented as the average macro value ($\pm$ standard deviation). The score with the greatest average value in each column is bolded. N/R means Not Reported.

Authors, year	Model performance				
	F_1	F_{1n}	F_{1a}	F_{1o}	F_{1p}
Liu et al. 2018 [33]	.840 ($\pm$ N/R)*	.903 ($\pm$ N/R)	.864 ($\pm$ N/R)	.751 ($\pm$ N/R)	.738 ($\pm$ N/R)
Rubin et al. 2018 [34]	.820 ($\pm$ N/R)*	.910 ($\pm$ N/R)	.830 ($\pm$ N/R)	.720 ($\pm$ N/R)	N/R
Fan et al. 2020 [35]	.851 ($\pm$.010)*	.926 ($\pm$.005)	.882 ($\pm$.010)	.745 ($\pm$.015)	N/R
Zhao et al. 2020 [36]	.802 ($\pm$ N/R)*	.893 ($\pm$ N/R)	.792 ($\pm$ N/R)	.723 ($\pm$ N/R)	.525 ($\pm$ N/R)
Cao et al. 2020 [37]	.820 ($\pm$ N/R)*	.910 ($\pm$ N/R)	.840 ($\pm$ N/R)	.700 ($\pm$ N/R)	N/R
Quang et al. 2021 [38]	.837 ($\pm$ N/R)*	.930 ($\pm$N/R)	.787 ($\pm$ N/R)	.793 ($\pm$N/R)	N/R
Weimann et al. 2021 [29]	.779 ($\pm$.014)	.915 ($\pm$.007)	.777 ($\pm$.014)	.763 ($\pm$.014)	.661 ($\pm$.040)
DenseNet-37	.791 ($\pm$.009)	.916 ($\pm$.003)	.799 ($\pm$.020)	.779 ($\pm$.011)	.668 ($\pm$.024)

^a The character (*) denotes the F_1 score that is averaged over three classes including normal rhythm (F_{1n}), AF rhythm (F_{1a}), and other rhythm (F_{1o}).

Table 4

Comparison of the number of parameters and F_1 score achieved by different DenseNet architectures for the ICBEB2018 dataset [9]. The score in each cell is presented as average macro value ($\pm$ standard deviation). The score with the greatest average value in each row is bolded.

Model performance	Model structure			
	DenseNet-37	DenseNet-121	DenseNet-169	DenseNet-201
F_1	.826 ($\pm$.012)	.799 ($\pm$.013)	.794 ($\pm$.012)	.790 ($\pm$.015)
F_{af}	.900 ($\pm$.011)	.885 ($\pm$.009)	.884 ($\pm$.007)	.872 ($\pm$.015)
F_{block}	.829 ($\pm$.019)	.811 ($\pm$.017)	.808 ($\pm$.010)	.808 ($\pm$.010)
F_{pc}	.796 ($\pm$.009)	.772 ($\pm$.025)	.772 ($\pm$.019)	.755 ($\pm$.028)
F_{st}	.877 ($\pm$.004)	.865 ($\pm$.009)	.869 ($\pm$.006)	.861 ($\pm$.008)
No. Parameters	68,177	5,617,289	10,638,601	15,926,793

the least number of parameters. We assume that DenseNet-121, -169, and -201 models (with a high number of convolution layers in general) are too complicated and overfit the training data.

The validation performance in terms of F_1 of each DenseNets architectures is depicted in Fig. 3. We deploy the early-stopping technique to automatically exit training iteration when the validation F_1 value stops increasing for 30 epochs. As shown in this figure, all DenseNets quickly achieve a high validation F_1 within just less than 130 epochs. The training procedure is repeated 10 times, the training and validation sets randomly vary from one to another within the pool of 80% recordings. We report the average macro validation F_1 as the dark color lines as well as the standard deviation by the light color regions. The validation F_1 curve of the proposed DenseNet-37 structure is shown to vary less and reach a higher value compared to other DenseNets models.

The overall F_1 scores achieved by the DenseNets architectures are presented in Fig. 4 in form of boxplot. It is obviously shown that the proposed DenseNet-37 model outperforms all the others with the highest median value.

Now we compare the performance of DenseNet-37 model with other existing works based on four sub-abnormal types of the ICBEB2018 dataset. The comparison is presented in Table 5, we look at the following metrics: F_1 , F_{af} , F_{block} , F_{pc} and F_{st} . The result shows that the DenseNet-37 performs relatively well compared to other works. Remarkably, it outperforms other models in terms of F_{st} metric. Our F_{st} value is 0.877 while no other studies reach 0.8 (except for the Refs. [39–41] as this value is not reported in works). It is worth recalling that when working with the ICBEB2018 dataset, the sampling frequency of the signal was down-sampled from 500 Hz to 250 Hz, and the signal length only extended to 30 s due to Google Colab Pro's limited RAM capacity. When doing so, we miss some information in the input, and that slightly reduces the performance of our model. We believe that with more RAM capacity, the performance of our model will be significantly improved.

Besides the above F_1 metrics, the classification performance of DenseNet-37 was also investigated on four other metrics for the ICBEB2018 dataset. In Table 6, we compare the results of the proposed

model with the original DenseNets architectures as well as the related works based on these indexes. Compared to DenseNet-121, -169 and -201, DenseNet-37 obtains the highest value for all metrics. Compared to other related works, DenseNet-37 model is slightly better than the other two models for metric F_{max} . However, the model proposed by Strodthoff et al. [30], which uses the convolutional neural networks, in particular ResNet- in combination with Inception-based architectures, is better for the other three metrics. The better results in the literature [29,30] can be explained by the fact that our input data was shorter (30 s versus 60 s) and at a lower down-sampling frequency (250 Hz versus 500 Hz). Additionally, in these works the model was pretrained on large datasets (Icentia11K and PTB-XL), and the transfer learning technique was used to classify the ECG signals.

To visually analyze the performance of the proposed DenseNet-37 model, we show in Fig. 5 the right attention mapping analysis for some classes using Grad-CAM [43]. From top to bottom are normal, atrial fibrillation (AF), ST-segment elevation (STE), and left bundle branch block (LBBB) ECG signals. The color mapping is shown on the right of each figure; the dark blue color presents a high concentration while the red color presents a low component of the corresponding ECG signal. Fig. 5a shows that in the normal ECG signal, the attention focuses on the T-waves. From Fig. 5b, one can see that the attention concentrates on abnormal P-waves for AF conditions. As shown in Fig. 5c, on ST-segment changed conditions, the attention is mainly focused on ST-segment. It is shown in Fig. 5d that the attention is paid to abnormal P-waves and QRS complex waves. In all figures, the predicted label is identical to the true label with an impressively high possibility (from 0.95437 to 1.00). This high agreement between the predicted result and the ground truth implies a good performance of the model and that it can be effectively applied for automatic ECG classification.

4. Conclusion

We proposed a lightweight model architecture based on the Densely Connected Convolutional Networks in combination with Cyclical Learning Rate to automatically classify ECG signals. The model was trained and evaluated on two freely accessible datasets, including CinC2017 and ICBEB2018. It was shown that the model with 37 convolution layers along with some adjustments worked better than their original DenseNet-121, -169, and -201 counterparts. We also proved that the proposed architecture performs best with the mini-batch size of 8. The metrics, F_1 for CinC2017 dataset and F_1 , AUC , F_{max} , $F_{\beta=2}$, $G_{\beta=2}$ for ICBEB2018 dataset, were utilized to compare our results with other works. Using DenseNet-37, we obtained the average macro F_1 score of 0.831 and 0.826 for CinC2017 and ICBEB2018 datasets, respectively. These results are in line with the state-of-the-art deep neural networks despite the fact that our model consists of a very small number of parameters. For further development, we believe that applying transfer learning technique to utilize the commonalities between different ECGs to our lightweight model will significantly improve classification performance.

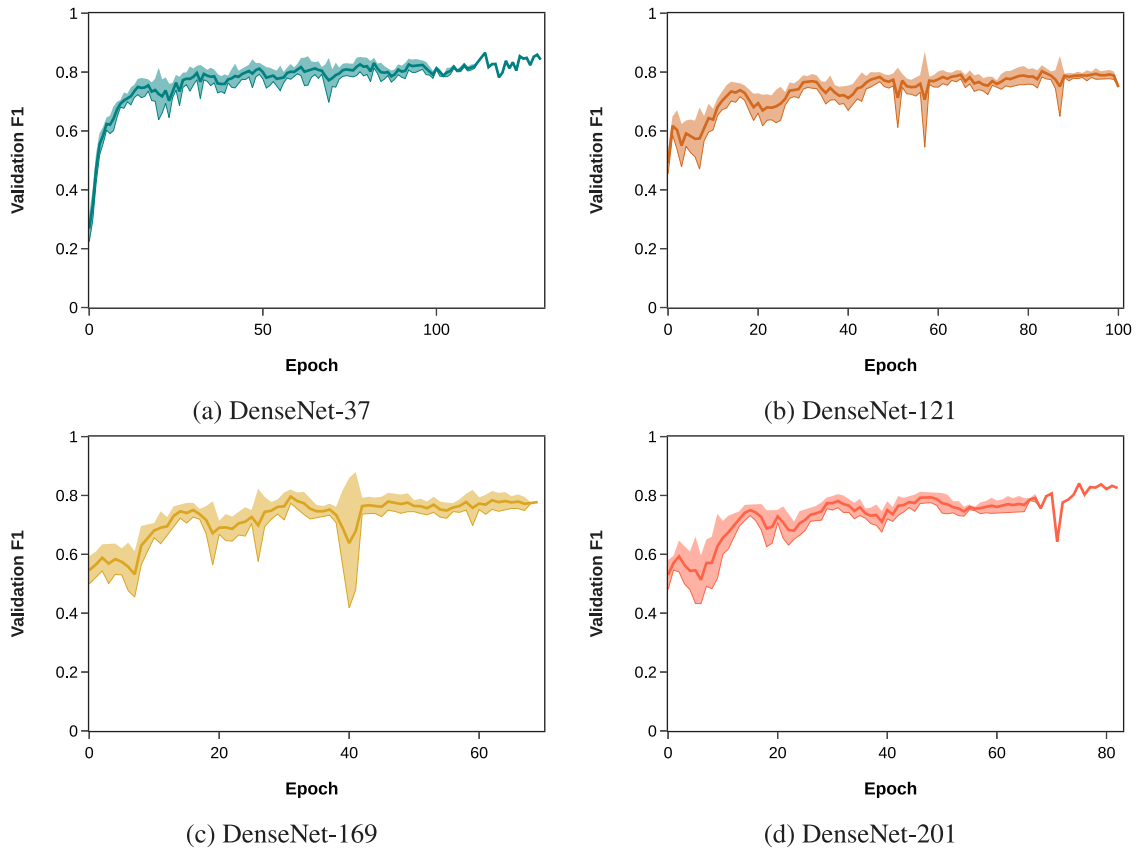


Fig. 3. Average macro F_1 score on the validation set during training. The dark color lines and light color regions present the average validation F_1 and the standard deviation, respectively.

Table 5

Comparison of the proposed network on our test set with related works for the ICBE2018 dataset [9]. The score in each cell is presented as average macro value ($\pm$ standard deviation). The score with the greatest average value in each column is bolded. N/R means Not Reported.

Authors, year	Model performance				
	F_1	F_{af}	F_{block}	F_{pc}	F_{st}
He et al. 2019 [42]	.806 ($\pm$ N/R)	.914 ($\pm$ N/R)	.879 ($\pm$ N/R)	.801 ($\pm$ N/R)	.742 ($\pm$ N/R)
Yao et al. 2020 [40]	.812 ($\pm$ N/R)	N/R	N/R	N/R	N/R
Chen et al. 2020 [10]	.837 ($\pm$ N/R)	.933 ($\pm$ N/R)	.899 ($\pm$ N/R)	.847 ($\pm$ N/R)	.779 ($\pm$ N/R)
Jeong et al. 2021 [41]	.740 ($\pm$ N/R)	N/R	N/R	N/R	N/R
Qin et al. 2021 [39]	.831 ($\pm$ N/R)	N/R	N/R	N/R	N/R
DenseNet-37	.826 ($\pm$.012)	.900 ($\pm$.011)	.829 ($\pm$.019)	.796 ($\pm$.009)	.877 ($\pm$.004)

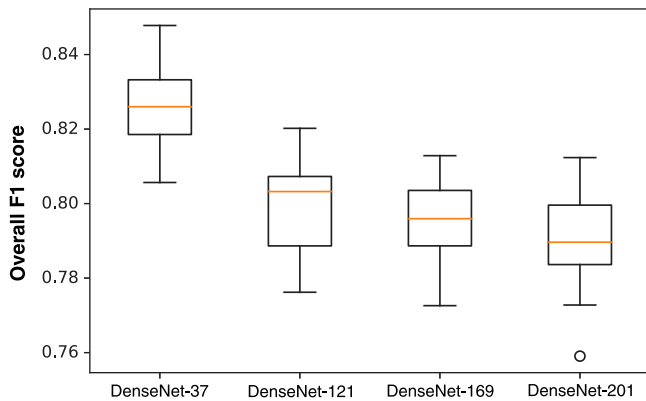


Fig. 4. The overall F_1 scores of four models in experiments. The empty circle presents the outlier.

Table 6

The proposed model performance with several training methods based on the different four metrics for the ICBE2018 dataset [9]. The score in each cell is presented as the average macro value ($\pm$ standard deviation). The score with the greatest average value in each column is bolded.

Model structure	Model performance			
	AUC	F_{max}	$F_{\beta=2}$	$G_{\beta=2}$
DenseNet-37	.969 ($\pm$.002)	.860 ($\pm$.005)	.804 ($\pm$.010)	.594 ($\pm$.011)
DenseNet-121	.962 ($\pm$.004)	.849 ($\pm$.005)	.773 ($\pm$.017)	.571 ($\pm$.017)
DenseNet-169	.959 ($\pm$.005)	.847 ($\pm$.010)	.791 ($\pm$.010)	.579 ($\pm$.017)
DenseNet-201	.960 ($\pm$.006)	.843 ($\pm$.007)	.777 ($\pm$.014)	.568 ($\pm$.013)
Related works				
Strodthoff et al. 2020 [30]	.974 ($\pm$.005)	.855 ($\pm$.020)	.819 ($\pm$.028)	.602 ($\pm$.044)
Weimann et al. 2021 [29]	.961 ($\pm$.004)	.854 ($\pm$.005)	.814 ($\pm$.008)	.591 ($\pm$.012)

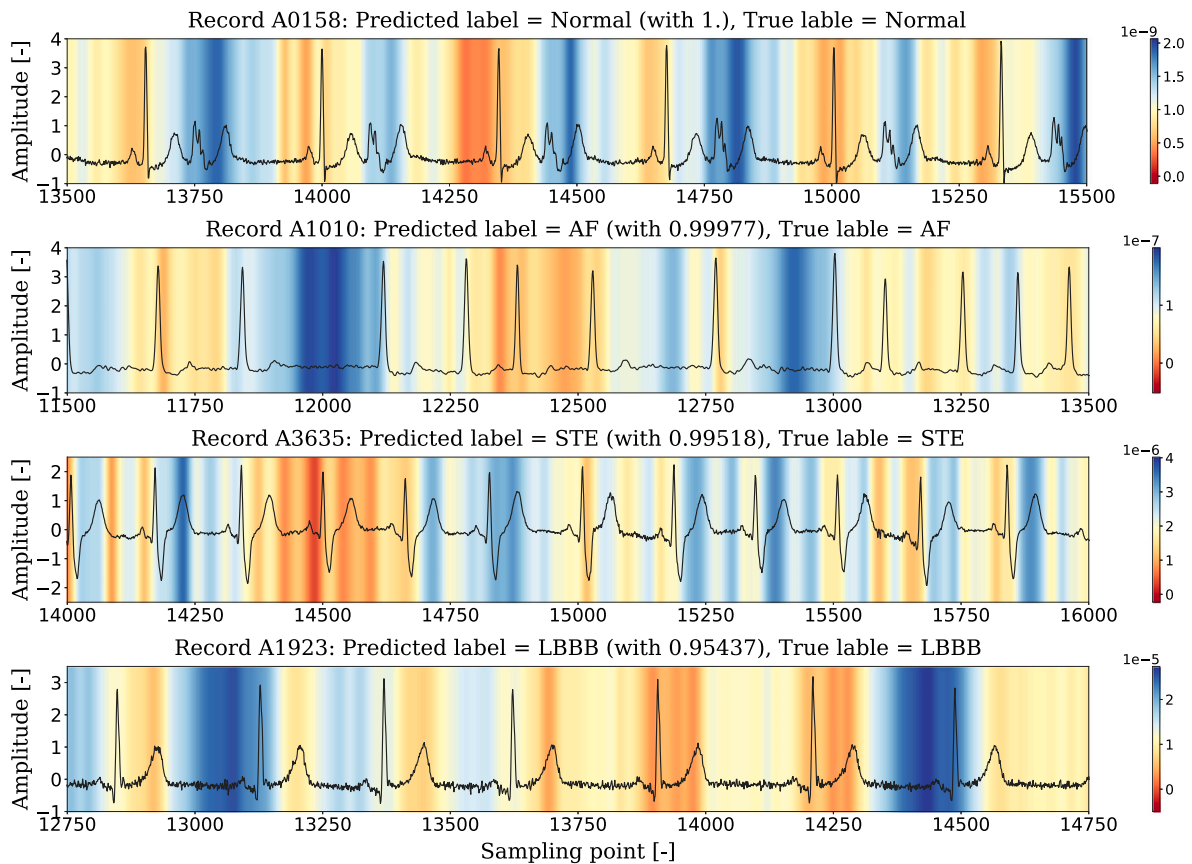


Fig. 5. Original ECG signal with attention mapping: a) Normal; b) Atrial fibrillation (AF); c) ST-segment elevated (STE); d) Left bundle brunch block (LBBB).

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CRediT authorship contribution statement

Bui Thi Hanh: Methodology, Software, Validation, Writing, Involved in its revision. **Hoang Van Manh:** Conceptualization, Investigation, Formal analysis, Involved in its revision. **Pham Manh Thang:** Conceptualization, Investigation, Formal analysis, Involved in its revision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.bspc.2022.104529>.

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